

Applied Performance Task, Expectations and Rubric

The Task

Students are to investigate the areas of mineral exploration in Canada. They are to create a poster using their ArcView Map.

Not a day passes without each one of us using objects made of minerals and metals. You will investigate where in Canada the minerals and the metals of these objects originate. Each student will have to find 15 pictures of objects around and in our home and identify which minerals and/or metals this specific object is made of. Using their ArcView Map and the web site "Minerals and Metals – A World to Discover" (http://www.nrcan.gc.ca/mms/school/e_mine.htm), students are:

- to identify the mineral and metal content of these 15 objects,
- to identify the different categories these minerals or metals belong to,
- to locate the ecozone from which each of the minerals and metals contained within these 15 objects would most likely come from.

They also will have to choose their object so that they would represent at least a minimum of 7 different ecozones and seven different minerals or metals.

Expectations

This task gives students the opportunity to demonstrate achievement of the following selected expectations from three strands: Geographic Foundations: Space and Systems; Human-Environment Interactions; and Methods of Geographic Inquiry.

Students will:

1. demonstrate an understanding of how natural and human systems interact within ecozones;
2. analyse ways in which Canadians use resources in Canada;
3. demonstrate an understanding of how natural systems (e.g., climate, soils, landforms, wildlife) influence cultural and economic activities (e.g., recreation, transportation, employment opportunities);
4. demonstrate an understanding of the methods used to collect, organize, manipulate, and interpret geographic data;
5. use geographic material from primary sources (e.g., field research, surveys, interviews) and secondary sources (e.g., mainstream and alternative media, CD-ROMs, Internet) effectively and appropriately when researching a geographic issue;
6. select and use appropriate methods for displaying geographic data;
7. use appropriate technology (e.g., computer maps, graphs, air photos) to present geographic information;
8. collect, organize, and synthesize information about the characteristics of Canada's geography from a variety of sources (e.g., atlases, photographs, hypermedia);
9. use cartographic conventions correctly when constructing maps (e.g., scale, legend, direction);
10. use graphic organizers (e.g. semantic webs, timelines, future wheels, analogy charts, Venn diagrams) effectively to visualize, clarify, and interpret geographic information.

Prior Knowledge and Skills

To complete this task, students are expected to be able to demonstrate the following:

- an understanding of the rock cycle



- explain the importance of mining in Canada
- describe the steps used by mining companies to discover and develop an ore body
- explain some issues facing Canada's mining industry
- an understanding of research methods using a variety of source materials
- ability to use graphic organizers
- ability to read, analyse, and interpret graphs and maps
- ability to use a variety of tools and technologies for research.

Task Rubric – Minerals and Metals around and in our home Poster						
Expectations	Criteria	Level 1 (50–59)	Level 2 (60–69)	Level 3 (70–79)	Level 4 (80–100)	
Knowledge / understanding						
1,3,4	- Demonstrates an understanding of how natural and human systems interact within ecozones - Demonstrates an understanding of how natural systems influence cultural and economic activities - Demonstrates an understanding of the methods used to collect, organize, manipulate, and interpret geographic data	- Demonstrates limited understanding of how natural and human systems interact within ecozones - Demonstrates limited understanding of how natural systems influence cultural and economic activities - Demonstrates limited understanding of the methods used to collect, organize, manipulate, and interpret geographic data	- Demonstrates some knowledge of how natural and human systems interact within ecozones - Demonstrates some understanding of how natural systems influence cultural and economic activities - Demonstrates some understanding of the methods used to collect, organize, manipulate, and interpret geographic data	- Demonstrates considerable knowledge of how natural and human systems interact within ecozones - Demonstrates considerable understanding of how natural systems influence cultural and economic activities - Demonstrates considerable understanding of the methods used to collect, organize, manipulate, and interpret geographic data	- Demonstrates thorough knowledge of how natural and human systems interact within ecozones - Demonstrates thorough understanding of how natural systems influence cultural and economic activities - Demonstrates thorough understanding of the methods used to collect, organize, manipulate, and interpret geographic data	
Thinking / Inquiry						
5,7,8	- Synthesizes information relevant to the theme - Makes connections between illustrations, minerals and metals, and ecozone location	- Synthesizes information with limited effectiveness - Makes connections with limited effectiveness	- Synthesizes information with some effectiveness - Makes connections with some effectiveness	- Synthesizes information with considerable effectiveness - Makes connections with considerable effectiveness	- Synthesizes information with a high degree of effectiveness - Makes connections with a high degree of effectiveness	
Communication						
6,9	- Uses visuals to illustrate the importance of mineral exploration in Canada - Communicates using appropriate methods and technologies through logical organization	- Uses visuals to illustrate the importance to a limited degree - Minimal logic in plan and sequence where organization interferes with comprehension	- Uses visuals to illustrate the importance to some degree - Some logic in plan and sequence where lapses of organization interfere with comprehension	- Uses visuals to illustrate the importance to a considerable degree - Logic in plan and sequence where organization is effective and does not interfere with comprehension	- Uses visuals to illustrate the importance to a high degree - Thoughtful and logic in plan and sequence where organization enhances comprehension	
Application						
2	Analyses ways in which Canadians use resources	Shows limited analyses	Shows some analyses	Shows considerable analyses	Shows thorough analyses	